

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.001160775833379789
rotor delay [s]: 0.154705736141
gimbal delay [s]: 0.040119102824
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 4.277600267842443e-17
common-scale FD null norm: 1.0742617969714812e-11

Dimensionless coordinate

log_mass_scale	-0.387960361064
log_second_moment_mode_xx	-2.30541172481
log_second_moment_mode_yy	0.799907106797
log_second_moment_mode_zz	4.1151658558
log_second_moment_mode_xy	-0.97976620476
log_second_moment_mode_xz	1.31475125575
log_second_moment_mode_yz	0.414144219305
dimensionless_cog_displacement_x	-0.0881515128413
dimensionless_cog_displacement_y	0.117129801863
dimensionless_cog_displacement_z	-0.246380760918
log_force_effectiveness_1	-0.566181332638
log_force_effectiveness_2	-0.647652597962
log_force_effectiveness_3	-0.566142178724
log_force_effectiveness_4	-0.441724767393

Physical representative

mass [kg]: 1.5954160713386178
CoG body [m]: [-0.01891322 0.02240979 -0.03769311]
force effectiveness: [0.56768912 0.52327267 0.56771135 0.64292657]
inertia [kg m²]:
[0.2063207448, -0.0007607051854, -0.05207973768]
[-0.0007607051854, 0.1270646788, -0.02176072394]
[-0.05207973768, -0.02176072394, 0.2669928015]
principal moments [kg m²]: [0.12264142 0.17880799 0.29892882]

Per-bag raw residual-wrench RMS

failure_1_single: [0.37611519 0.12746613 0.57319249 0.06024084 0.11122275 0.09798997]